Brian Zheng

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Technical Skills

Languages: Java, Python, C/C++, JavaScript, HTML/CSS
Libraries: GStreamer, PyTorch, TensorRT, OpenCV, OpenGL, CUDA, WebRTC
Developer Tools: ROS2, Docker, Isaac Sim, AWS EC2, Foxglove, Git, SOLIDWORKS, Linux/Ubuntu

Experience

enVgo — Autonomy Software Developer Sept 2025 – Present
Waterloo, ON

- Boosted segmentation model accuracy by **19%** via dataset refinement and synthetic data generation, while concurrently reducing inference latency on **Jetson** edge hardware using **TensorRT** and **CUDA**.
- Developed real-time **Bird's-Eye View** through **homography transformations** using **GStreamer/OpenGL** to stitch 4 fisheye camera feeds, optimized through **GPU Kernels** and Jetson **hardware encoders**.
- Developed a **deep RL agent** to suppress radar returns by learning policies over **Doppler coherence**, and an **EKF** to fuse consistent measurements into stable tracked point clouds.
- Developed quantitative benchmarking for segmentation models through Dockerized metric scripts, enabling data-driven model comparison via standardized **mIoU metrics**.
- Used **NVIDIA Isaac Sim** to develop a realistic marine environment with multi-sensor rendering (RGB, depth), radar ray-tracing models, hydrodynamic wave physics, and vessel dynamics for testing perception pipelines.

WATonomous — Team Captain & Director of Vehicle Autonomy Jan 2025 – Present
University of Waterloo

- Leading a team of **20+** members to develop a **Level 4** autonomous vehicle and a **full-stack perception system**, integrating sensor fusion and **deep learning/ML** based 3D object detection and tracking.
- Developed a hybrid **3D tracking pipeline** by fusing **LiDAR** clustering (DBSCAN) with camera-based **YOLOv8** detections, accelerating inference via **TensorRT** quantization.
- Architected a custom **JDE network** based on **YOLOv8**, adding a dual-head design for joint **detection and ReID embeddings**, enabling single-pass tracking inference.
- Developed and fine-tuned a **BEVFusion** model with **LiDAR** and **camera feature encoders**, BEV view transformation, and anchor-free center **heatmap/regression heads** for **3D object detection** with class-aware bounding boxes.
- Contributed to **ROS2** open-source buildfarm called **DeepROS**, developing model-agnostic node containers for easy optimized deployment of ML models.

Projects

End-to-End Autonomous Drone | *ROS2, TensorFlow, AirSim, ArduPilot, PX4, YOLOv4*

- Engineered a **ROS2-based** navigation stack integrating **Smac Planner** for dynamic 3D pathfinding and **Stereo-based voxel grids** to navigate zones and static obstacles.
- Developed custom object detection models using **YOLOv4-tiny** and integrated them into a real-time drone control pipeline via **ROS2** and **MAVSDK**.
- Validated mission architecture via **SITL** simulations in **AirSim**, achieving autonomous payload delivery tasks to **PX4** hardware.

Alpha-Zero Engine (ChessHacks 2025 Finalist) | *Python, ResNet, CUDA, Pytorch*

- Developed an **end-to-end chess engine** using **PyTorch/CUDA**, designing a dual-head **ResNet** to map 18-plane board representations to 4,672 policy logits and value estimates
- Integrated **MCTS/PUCT** search guided by **neural network priors** and **value estimates**; used **transposition tables**, **FP16 LRU caching**, and **batched leaf evaluations** to achieve **3–5×** throughput and **<200 ms** typical move latency.
- Improved model accuracy using **RL self-play** and **DAgger** distillation to refine policy/value network using Stockfish; implemented a rolling replay buffer, mitigating **distribution shift** while optimizing via **KL-Divergence** and MSE losses.

Education and Certifications

University of Waterloo
Bachelor of Applied Science, BASc, Mechatronics, Robotics, and Automation Engineering *Aug 2024 – May 2029*